Week 6 — Reading Pack

Forecasting in Pharma: the launch record, the patient journey as a funnel, epidemiology, persistence, the LOE calendar, and the core readings

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Read before Monday, October 19 · ~75 minutes · one written reflection (any of A–E) to bring to class

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Sources are limited to peer-reviewed papers, standard texts, government agencies, and the IQVIA papers in the course library. Industry passages are quoted verbatim from the IQVIA PDFs in this folder; open-access papers are saved in sources/ (run download_sources.sh); the rest are listed with DOIs in Part F. Each section ends with a reflection; write one, bring it.

A. How to read this pack

Tonight's spine is the patient funnel — a simple chain of documented assumptions, read at two horizons (budget and long-range plan). The readings supply what the synthetic data cannot: the industry's record against its own forecasts (B), real attrition in a real patient journey (C), why epidemiology is the first number and rarely the most uncertain one (C2), what the patent calendar looks like (D), and the core texts (E). Part F lists every source with what it is used for.

B. The launch record: 50–70% miss (objectives 1 & 3)

IQVIA's launch-excellence work is the industry's own audit of its forecasts. The passages below are the reason tonight insists on an assumption book, a range, and a bias log.

> *In studying about 12,000 product launches across eight countries and 10 years, one of the most important factors is setting expectations. Between 50% and 70% of launch brands underperform their first-year forecast. That suggests companies routinely create unrealistic expectations or fail to understand the opportunity.*
>
> — IQVIA, "A Framework for Successful Biopharma Launches" (Gupta & McClellan), Critical Success Factors

> *A brand's fate is sealed before the launch even begins. The quality of pre-launch preparation, resources, alignment and processes determines post-launch success.*
>
> — IQVIA, "A Framework for Successful Biopharma Launches," same section

> *IQVIA analyzed more than 600 launches based on two axes of unmet need and product differentiation. What they found was four archetypes with distinct critical success factors and distinct first-year sales … The upper right quadrant (high unmet need and high product differentiation) is labeled Science Sells. Products in this area have the highest average first-year sales: $94 million. The other end of the spectrum, low unmet need and low product differentiation, is Who Benefits, with average first-year sales of $19 million.*
>
> — IQVIA, "A Framework for Successful Biopharma Launches," The Framework

> *That was the position Amarin found itself in when launching Vascepa®. The product, a fish oil to treat severely high triglycerides, was launched in early 2013. From the company perspective, Vascepa had an important distinction: it lowered triglycerides without increasing LDL-cholesterol. The mechanism of action inspired revenue estimates of $60 million for 2013 and $1.9 billion in peak sales for 2022. Prescribers did*

> *not agree. … Vascepa generated only $32 million in first-year sales, less than half of Lovaza's first-year sales.*
>
> — IQVIA, "A Framework for Successful Biopharma Launches," Who Benefits

Read the Vascepa paragraph as a forecast post-mortem. The $60M / $1.9B gap was a share assumption — 'the mechanism will persuade prescribers' — applied to a large pool. The pool was right; the share was wrong by a factor of two in year one. Tonight's assumption book writes peak share down with a low, a high and a reason. The archetype table is a starting point for that range: a 'Who Benefits' product does not get a 'Science Sells' share.

Reflection B. LYMBREXA is a third-in-class product with a safety and convenience advantage (subcutaneous, fixed duration, outpatient step-up) in a high-unmet-need setting. Place it in IQVIA's 2×2. Then write three sentences on what the archetype implies for p, q and m relative to the first-in-class analog — and which of the three the brand team is most likely to overstate.

C. The patient journey as a funnel: the CAR T waterfall (objective 1)

A funnel is only as good as its attrition rates, and the industry's best-documented attrition is IQVIA's CAR T-cell therapy journey in large B-cell lymphoma — the same disease as tonight's case. The numbers below are why cell-therapy analogs ramp slowest, and why 'eligible' and 'reached' are different stages.

> *In the United States, most patients begin their treatment journey at a practice likely located near their home. They are often managed by a general oncologist or hematologic oncologist at an institution without advanced CAR T-cell treatment capabilities.*
>
> — IQVIA, "The Path to CAR T-cell Therapy" (2025), Introduction

> *This research uncovered a substantial disparity between the number of patients who may be eligible for the treatment and those who ultimately receive it. … 63% of eligible patients are never referred to a treatment center … 16% of eligible patients ultimately receive CAR T-cell therapy treatment.*
>
> — IQVIA, "The Path to CAR T-cell Therapy," Introduction and headline figures

> *The findings revealed that almost two-thirds of eligible patients never progress through this initial stage. Moreover, the research indicated that overall, less than 2 out of 10 LBCL patients eligible for CAR T-cell therapy ultimately receive the treatment.*
>
> — IQVIA, "The Path to CAR T-cell Therapy," The CAR T therapy waterfall

> *Approximately 30% of the drop-off that occurs while in the referring setting can be attributed to perceived patient ineligibility. … Oncologists who have less comfort with CAR T-cell therapy, however, tend to apply lower age cutoffs and are more likely to refer only younger patients (<70 years of age). It is also important to note that FDA labels do not specify an upper age limit.*
>
> — IQVIA, "The Path to CAR T-cell Therapy," Barriers experienced at a referring center

> *IQVIA's 2024 LBCL research found that 46% of surveyed referring oncologists used ECOG ≥ 2 as a cutoff threshold for CAR T referral eligibility, while 38% used ECOG ≥ 3.*
>
> — IQVIA, "The Path to CAR T-cell Therapy," same section

Map it onto tonight's funnel. 'Eligible' is the 2L / 3L+ starts the funnel produces. 'Referred' and 'treated' sit in the share and coverage assumptions — for a cell therapy $0.37 \times 0.43 \approx 0.16$ before any competition. A subcutaneous bispecific given in the community skips most of this waterfall, which is why its year-1 uptake assumption is closer to the outpatient analogs than to CAR-T.

Referral also takes time, and the 'reached' pool is what prescriber concentration describes from the other side: a few hundred prescribers at treatment centers write most of the starts — the basis of the coverage assumption.

Reflection C. Rewrite IQVIA's waterfall as four rows of an assumption book for a cell-therapy launch: parameter, low / base / high, source, and the evidence that would move it. Then say which single row a subcutaneous bispecific can delete.

C2. Epidemiology: the first number, rarely the most uncertain one (objective 1)

> *Diffuse large B-cell lymphoma (DLBCL) is the most common NHL subtype, comprising approximately 30%–40% of NHL cases, with incidence rising with age.*
>
> — IQVIA, "Self-Driving CAR for Hematologic Malignancies," Introduction

> *Over recent decades, the global incidence of HMs has steadily increased, while age-adjusted mortality rates have declined, largely due to advances in treatment.*
>
> — IQVIA, "Self-Driving CAR for Hematologic Malignancies," same section

> *Rare disease innovators must understand and evidence disease epidemiology, starting with accurate incidence and prevalence figures. Understanding true prevalence and incidence figures can be vital in making the case for health system priorities, as orphan medicines are generally perceived as high cost, to the point where treating 50 vs. 100 makes a major difference.*
>
> — IQVIA, "From Orphan to Opportunity: Mastering Rare Disease Launch Excellence," Value and evidence

> *Accurate patient numbers reassure payers and allow budget planning for swifter access. RWE generation helps insight on complex disease biology and the natural history of the disease and can lay the basis for better patient finding.*
>
> — IQVIA, "From Orphan to Opportunity," same section

Two peer-reviewed anchors for the DLBCL funnel. Sehn & Salles (NEJM 2021): more than 60% of patients are cured with first-line R-CHOP, so roughly a third or more relapse or are refractory — the basis of tonight's 1L → 2L assumption. Teras et al. (CA Cancer J Clin 2016): DLBCL is the most common US lymphoid subtype, and incidence rises steeply with age.

Carry one thing into class: the orphan paper's '50 vs. 100' is the rare-disease case where incidence is the widest bar of the tornado; in DLBCL it is among the narrowest, and share and months on therapy are the widest. The tornado tells you which situation you are in.

Reflection C2. For Homework 2's disease (mantle cell lymphoma, ~0.8 per 100,000) decide, before you run anything, whether you expect the tornado to look like DLBCL's or like a rare disease's — and write down the two funnel parameters you would buy evidence on and from which source.

D. The LOE calendar (objective 3)

> *Polivy will maintain its exclusivity in Europe until January 2028 and in the USA until June 2031. Its USA molecule patent will lose its protection in August 2029, while the European molecule patent, extended by SPC, will remain valid until July 2033.*
>
> — IQVIA, "ADCs in Cancer" (Ark Patent Intelligence, September 2025), Table 3 discussion

A forecaster reads three dates for every molecule — compound patent, regulatory exclusivity, pediatric or SPC extension — and the later one governs. The erosion after that date is borrowed from other drugs' history. The

peer-reviewed benchmark: one year after generic entry, U.S. brands kept on average 23% of their unit volume (18% for drugs with more than $250M in sales; Grabowski et al., J Med Econ 2021). Tonight's GLUCOREN example applies that number to the brand's trend.

Reflection D. GLUCOREN's compound patent expires July 2027. Write three rows of an assumption book for the patent expiry (expiry date, share kept in the expiry year, share kept after) with low / base / high and a source or reason for each.

E. Core readings — a guided walk

E1. Cook (2015), Forecasting for the Pharmaceutical Industry (2nd ed.) — the new-product chapters

The standard practitioner text. Read the new-product forecast chapters ('Modeling the market', 'Forecasting the product', 'Converting patient to revenue') for the patient-based logic of tonight's funnel. Note how the book separates the market model from the product model — the same split as tonight's funnel and brand share.

E2. Briggs & Sculpher (1998) — flow models in health care (optional)

The classic introduction to cohort models that move patients between states over time. Background for the flow-vs-stock slide; tonight's funnel is the simplest possible flow model (one-year steps between lines).

E3. Clark et al. (2003) and Cramer et al. (2008) — survival and persistence

Clark et al. (open access) is a short tutorial on censoring, survival and hazard functions, and the Kaplan–Meier estimator — the method behind both the progression rates and the persistence curves. Cramer et al. is the ISPOR definition paper: persistence is the time from initiation to discontinuation; compliance (adherence) is taking the dose as prescribed. Use their terms in Homework 2.

E4. Kalyanaram, Robinson & Urban (1995); Bass (1969, optional)

Kalyanaram, Robinson & Urban summarize order-of-entry evidence: for packaged goods and prescription anti-ulcer drugs, an entrant's share relative to the pioneer is about 1/√(order of entry) — the basis of LYMBREXA's 15% peak share. Bass is the formal S-curve of adoption (innovators and imitators); tonight replaces it with a simpler uptake table read from analog launches.

E5. Hyndman & Athanasopoulos, Forecasting: Principles and Practice (3rd ed.) — ch. 8 and 11.1

- **Ch. 8:** exponential smoothing — the market-trend check.
- **Ch. 11.1:** hierarchical series; read only the top-down approach used tonight for the regional split.

E6. Ansari et al. (2024), "Chronos: Learning the Language of Time Series" — §1–3 and the zero-shot results

A time-series foundation model built as language modeling: scale the series, turn values into tokens, train a T5-family model on a large corpus, forecast by sampling. Read §1 for the framing and the benchmark tables. TimesFM (Das et al., ICML 2024) is a peer-reviewed alternative; Lag-Llama is a preprint; TimeGPT-1 is a vendor's description of a commercial model and should be read as such. All need history — none forecasts a launch.

Reflection E. Pick the LYMBREXA assumption you trust least. Write its row (low / base / high, source type, reason) and say what evidence, available within 90 days, would narrow it.

F. References

Course library = the IQVIA PDFs in this folder. 'sources/' = open-access copies saved by download_sources.sh. Everything else: follow the DOI (Columbia Libraries provides access).

- Cook, A. G. **Forecasting for the Pharmaceutical Industry: Models for New Product and In-Market Forecasting and How to Use Them (2nd ed.).** Gower / Routledge, 2015. https://www.routledge.com/9781032838571 — The standard practitioner text: new-product (patient-based) vs in-market (trend-based) forecasting, converting patients to revenue, how forecasts are used by stakeholders.
- Gupta, M. & McClellan, W. (IQVIA) **A Framework for Successful Biopharma Launches.** IQVIA white paper, 2017 — course library. a-framework-for-successful-biopharma-launches.pdf [course library] — "Between 50% and 70% of launch brands underperform their first-year forecast" (~12,000 launches) — why the assumption book and the range matter.
- Centers for Disease Control and Prevention **Principles of Epidemiology in Public Health Practice (3rd ed.), Lesson 3: Measures of Risk.** CDC, 2006 (updated 2012). https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson3/section2.html — Incidence vs prevalence: "Prevalence is based on both incidence and duration of illness."
- Teras, L. R., DeSantis, C. E., Cerhan, J. R., et al. **2016 US lymphoid malignancy statistics by World Health Organization subtypes.** CA: A Cancer Journal for Clinicians 66(6):443–459, 2016. https://doi.org/10.3322/caac.21357 — DLBCL is the most common lymphoid neoplasm subtype in the US; incidence rises steeply with age — the age-banded front of Funnel #1.
- IQVIA **Self-Driving CAR for Hematologic Malignancies.** IQVIA white paper, 2025 — course library. self-driving-car-for-hematologic-malignancies.pdf [course library] — DLBCL "comprising approximately 30%–40% of NHL cases, with incidence rising with age".
- Sehn, L. H. & Salles, G. **Diffuse Large B-Cell Lymphoma.** New England Journal of Medicine 384:842–858, 2021. https://pubmed.ncbi.nlm.nih.gov/33657296/ — More than 60% of patients are cured with first-line R-CHOP; the rest relapse or are refractory — the basis of assumption T2.
- IQVIA **The Path to CAR T-cell Therapy: Barriers in LBCL.** IQVIA white paper, 2025 — course library. iqvia-the-path-to-car-tcell-therapy-white-paper-2025.pdf [course library] — 63% of eligible LBCL patients never referred; 16% ultimately treated — why cell-therapy launches ramp slowly (the low uptake case).
- IQVIA **From Orphan to Opportunity: Mastering Rare Disease Launch Excellence.** IQVIA white paper, 2025 — course library. from-orphan-to-opportunity-mastering-rare-disease-launch-excellence.pdf [course library] — "Accurate incidence and prevalence figures … treating 50 vs. 100 makes a major difference" — the diagnosed-prevalence funnel.
- Kaplan, E. L. & Meier, P. **Nonparametric Estimation from Incomplete Observations.** Journal of the American Statistical Association 53:457–481, 1958. https://doi.org/10.1080/01621459.1958.10501452 — The Kaplan–Meier estimator for censored data — used to justify T2, T3 and D1.
- Clark, T. G., Bradburn, M. J., Love, S. B. & Altman, D. G. **Survival Analysis Part I: Basic concepts and first analyses.** British Journal of Cancer 89:232–238, 2003. https://doi.org/10.1038/sj.bjc.6601118 [sources/Clark_2003_Survival_Analysis_Part_I_BJC.pdf] — Open-access tutorial: censoring, survival and hazard functions, Kaplan–Meier.
- Briggs, A. & Sculpher, M. **An Introduction to Markov Modelling for Economic Evaluation.** PharmacoEconomics 13(4):397–409, 1998. https://doi.org/10.2165/00019053-199813040-00003 — The classic introduction to cohort (flow) models in health care — background for flow vs stock. Optional.
- Cramer, J. A., Roy, A., Burrell, A., et al. **Medication Compliance and Persistence: Terminology and Definitions.** Value in Health 11(1):44–47, 2008. https://doi.org/10.1111/j.1524-4733.2007.00213.x — ISPOR definitions: persistence = duration from initiation to discontinuation; compliance (adherence) = dose taken as prescribed.
- Little, J. D. C. **A Proof for the Queuing Formula: L = λW.** Operations Research 9(3):383–387, 1961. https://doi.org/10.1287/opre.9.3.383 — Little's law: patients on therapy = new patients × average months on therapy (in steady state).
- Bass, F. M. **A New Product Growth for Model Consumer Durables.** Management Science 15(5):215–227, 1969. https://doi.org/10.1287/mnsc.15.5.215 — The Bass diffusion model — the formal S-curve behind analog uptake ramps. Optional; tonight uses a ramp table instead.
- Kalyanaram, G., Robinson, W. T. & Urban, G. L. **Order of Market Entry: Established Empirical Generalizations, Emerging Empirical Generalizations, and Future Research.** Marketing Science 14(3):G212–G221, 1995. https://doi.org/10.1287/mksc.14.3.G212 — An

entrant's share ÷ the pioneer's ≈ 1 / √(order of entry) for packaged goods and prescription anti-ulcer drugs — the basis of assumption B3.

- Zipf, G. K. **Human Behavior and the Principle of Least Effort.** Addison-Wesley, 1949. https://archive.org/details/in.ernet.dli.2015.90211 — The rank-size law: a few prescribers write most of the prescriptions — sets coverage in the launch year.
- Centers for Medicare & Medicaid Services **Medicare Part D Prescribers — by Provider and Drug (public use file).** CMS, annual. https://data.cms.gov/provider-summary-by-type-of-service/medicare-part-d-prescribers — Real prescriber-level claim counts — students can check prescriber concentration (the rank-size pattern of the synthetic file) on real data.
- Grabowski, H., Long, G., Mortimer, R. & Bilginsoy, M. **Continuing trends in U.S. brand-name and generic drug competition.** Journal of Medical Economics 24(1):908–917, 2021. https://pubmed.ncbi.nlm.nih.gov/34253119/ — One year after generic entry, brands kept on average 23% of units (18% for drugs with > $250M sales) — the basis of assumption L3.
- IQVIA (Ark Patent Intelligence) **ADCs in Cancer.** IQVIA white paper, September 2025 — course library. iqvia-ark-pi-adc-cancer-whitepaper-09-25-forweb.pdf [course library] — Regulatory exclusivity vs molecule-patent dates — the LOE calendar.
- Hyndman, R. J. & Athanasopoulos, G. **Forecasting: Principles and Practice (3rd ed.).** OTexts, 2021 (free online). https://otexts.com/fpp3/ — Exponential smoothing (ch. 8) for the market check; top-down split (ch. 11). Syllabus reading.
- Ansari, A. F., Stella, L., Turkmen, C., et al. (Amazon) **Chronos: Learning the Language of Time Series.** Transactions on Machine Learning Research, 2024. https://arxiv.org/abs/2403.07815 [sources/Ansari_2024_Chronos_TMLR.pdf] — Tokenized time series + a T5-family model; zero-shot accuracy competitive with task-specific models on 42 benchmark datasets. Syllabus reading.
- Das, A., Kong, W., Sen, R. & Zhou, Y. (Google) **A decoder-only foundation model for time-series forecasting.** ICML 2024. https://arxiv.org/abs/2310.10688 [sources/Das_2024_TimesFM_ICML.pdf] — TimesFM, a decoder-only foundation model. Optional.
- Rasul, K., Ashok, A., Williams, A. R., et al. **Lag-Llama: Towards Foundation Models for Probabilistic Time Series Forecasting.** arXiv preprint, 2023. https://arxiv.org/abs/2310.08278 [sources/Rasul_2023_Lag-Llama_arXiv.pdf] — Lag-Llama — named in the syllabus; preprint. Optional.
- Garza, A., Challu, C. & Mergenthaler-Canseco, M. (Nixtla) **TimeGPT-1.** arXiv preprint, 2023 (vendor technical report). https://arxiv.org/abs/2310.03589 [sources/Garza_2023_TimeGPT-1_arXiv.pdf] — TimeGPT — named in the syllabus; a vendor's own description of a commercial model, read as such. Optional.

Bring to Week 6

One written reflection (any of B–E). Homework 2 is released tonight — its disease is in Reflection C2.